MIL-T-38049A(USAF)
AMENDMENT 2
15 APRIL 1983
Superseding
AMENDMENT 1
8 Feb 1978

MILITARY SPECIFICATION

TRANSMITTER, PRESSURE TRU-60/A

This amendment forms a part of Military Specification, MIL-T-38049A(USAF), dated 27 November 1972 and is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

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2.1: Under "SPECIFICATIONS, Federal" add:

"PPP-B-601 - Boxes, Wood, Cleated Plywood"

2.1: Under "SPECIFICATIONS, Military" delete "MIL-E-5272" and title thereto.

2.1: Under "STANDARDS, Military", add:

"MIL-STD-129 - Marking for Shipment and Storage

MIL-STD-810 - Environmental Test Methods"

Add the following boxed paragraph to the bottom of page:

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

PAGE 2

3.1: Delete paragraph and substitute the following:

"3.1 First article. When specified (see 6.4), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.4)."

PAGE 4

TABLE I, line 4, column 3: Delete "37.50" and substiutute "31.50".

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4.2,a: Delete "Qualification tests" and substitute "First article inspection".

FSC 6620

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4.4: Delete "Qualification testing" and substitute "First article inspection".

4.4.2, line 1: Delete "qualification tests" and substitute "first article inspection".

4.4.3, title and line 1: Delete "qualification tests" and substitute "first article inspection".

4.4.3.1, title and line 1: Delete "qualification test schedule" and substitute "First article inspection schedule".

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TABLE III, paragraph 4.6.8, under Test column, delete "overpressure" and substitute "sealing".

TABLE III, paragraph 4.6.9: Place an "X" in the Test Sample No. 1 and Test Sample No. 2 columns.

TABLE III, paragraph 4.6.20, under Test column, delete "Sealing" and substitute "Overpressure".

TABLE III, paragraph 4.6.21: Place an "X" in the Test Sample No. 2 and Test Sample No. 3 columns. TABLE III, add under "Transmitter Test Schedule": "NOTE" This table applies to first article inspection only."

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4.5.2.1,j: Delete "Overpressure" and substitute "Sealing".

4.5.2.2,i: Delete "Sealing" and substitute "Overpressure".

4.5.2.2,j: Delete entirely.

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4.6.3, line 5: Delete "25 ± 2°C" and substitute "23 ± 5°C".

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4.6.15: Delete paragraph and substitute the following:

"4.6.15 Vibration. The transmitter shall be tested within the range of frequencies and vibratory accelerations specified in MIL-STD-810, Method 514.2, procedure I, part 1, with vibration isolators, curve D, with the following changes: (1) Starting amplitude of 0.080 inch; (2) first break point at 10 Hz; (3) second break point at 15 Hz with amplitude of 0.036 inch; and (4) third break point at 75 Hz."

4.6.15.1: Delete paragraph entirely and substitute:

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"4.6.15.1 Resonant frequencies. Function of the transmitter shall be checked concurrently with the operation of resonance frequency search. This test shall be conducted at room temperature. During each resonance dwell period, the transmitter shall be connected to an indicator standard and subjected to a scale error test at the following pressure values. The tolerance for this test shall be 0.025 pressure ratio units.

a. Inlet pressures, inches Hg absolute: 8, 20 and 32.

b. Pressure ratios at each inlet pressure: 1.5 and 2.2."

4.6.15.1.1: Delete entirely and substitute:

"4.6.15.1.1 When resonant frequencies are not apparent within the specified frequency range, the transmitter shall be vibrated for a two-hour period with vibration isolators at a frequency of 55 Hz and an amplitude of 0.060 inch on each of three mutually perpendicular axes."

4.6.15.2, line 2: Delete "4.6" and substitute "4.6.2".

4.6.16: Delete entirely and substitute:

"4.6.16 Humidity. The transmitter, except when completely hermetically sealed, shall be subjected to the humidity test in accordance with MIL-STD-810, Method 507.1, procedure I, with the following changes: (1) Step 2: raise chamber temperature to 71°C (160°F); (2) Step 4: maintain 95 ± 5-3 percent humidity; and (3) Step 6: change 85% to 95 ± 5-3%".

4.6.17: Delete entirely and substitute:

"4.6.17 Fungus: The transmitter, except when completely hermetically sealed, shall be subjected to the fungus test in accordance with MIL-STD-810, Method 508.1, procedure I, for a period of 14 days."

4.6.18: Delete entirely and substitute:

"4.6.18 Salt fog. The transmitter shall be subjected to the salt for test in accordance with MIL-STD-810, Method 509.1, procedure I."

4.6.19: Delete entirely and substitute:

"4.6.19 Dust (fine sand). The transmitter shall be subjected to the dust test in accordance with MIL-STD-810, Method 510.1, procedure I, with the following changes: (1) Step 1: Adjust the air velocity to 300 ± 200 feet per minute; (2) Step 2: Maintain dust feed and air velocity at 300 ± 200 feet per minute. Raise the internal chamber air temperature to 71° C (160°F). Hold these conditions for 6 hours; (3) Step 3: Delete. After dust test, check transmitter accuracy at room temperature as specified in Table IV at the short scale error test points."

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4.6.21: Delete the two sentences beginning on line 20 and ending on line 24 and substitute "After completion of reliability testing, each transmitter shall be required to pass the individual tests listed in paragraph 4.5.1. The MTBF of the transmitter, which shall be at least 750 hours, shall be determined from the equation listed below."

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Delete Section 5 in its entirety and substitute the following:

"5. PACKAGING

5.1 Preservation - packaging. Preservation - packaging shall be level A or C as specified

5.1.1 Level A.

5.1.1.1 Cleaning Transmitters shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Transmitters shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation application. Not applicable.

5.1.1.4 Unit packaging. Unless otherwise specified by the procuring activity, each pressure transmitter TRU-60/A, shall be packaged in quantity unit packs of one each in accordance with method IA-8 of MIL-P-116. Each transmitter shall be overboxed in PPP-B-636 carton. Carton shall be large enough to allow for application of sufficient cushioning material between container and bag, of a type, density, and thickness to insure shock transmission does not exceed peak values in G's established for the transmitter when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Transmitters shall be clean, dry and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C as specified by procuring activity.

5.2.1 Level A. Items packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, styles A & B, Class overseas, unless otherwise specified by the procuring activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, and of minimum tare and cube, consistent with the protection required.

5.2.2 Level B. Items packaged as specified in 5.1.1 shall be packed in weather-resistant class exterior shipping container conforming to PPP-B-636. Other requirements as specified in 5.2.1 above are applicable.

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5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first domestic receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations and may be the contractor's commercial practice provided the latter meets the requirements of this level.

5.3 Marking. In addition to any special marking required by the contract or order, each unit package and shipping container shall be marked in accordance with MIL-STD-129."

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6.4: Delete entirely and substitute:

"6.4 First article. When a first article is required, the item will be inspected and tested in accordance with Section 4 of this document. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test and approval of the first article."

NOTE: Asterisks were not used at margins to indicate changes from previous amendment due to the extensiveness of the changes.

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